

Sputnik Light Package Insert R00 PKC038/21

Date : 29-12-2021

																				
Vector vaccine for the prevention of the coronavirus infection induced by the SARS-CoV-2 virus - (Sputnik Light) The drug is registered using the registration procedure for drugs intended for use in the situation of a threatened emergency or emergency response. The instruction is based on live subunit data on the use of the product and will be updated as new data becomes available. The product may only be administered in healthcare institutions authorized to handle the population under the established procedure.																				
1. GENERIC NAME COVID-19 Vaccine																				
2. QUALITATIVE AND QUANTITATIVE COMPOSITION <i>Composition per dose (0.5 ml):</i> Active substance: Recombinant adenovirus serotype 26 particles, containing the SARS-CoV-2 protein S gene, (1.0x10⁹ - 10¹⁰) particles per dose. Excipients: Trihydroxyethylamine(hydroxide) - 1.21 mg, sodium chloride -2.19 mg, sucrose - 25.0 mg, polyborate 80 - 250.0 mg, magnesium chloride hexahydrate - 102.0 µg, EDTA-disodium salt dihydrate -19.0 µg, ethanol 95% -2.51, water for injection Q.S. to 0.5 ml.																				
3. DOSAGE FORM AND STRENGTH A solution for intramuscular injection, 0.5 ml/ dose.																				
4. CLINICAL PARTICULARS 4.1 Indications: Sputnik Light Vector vaccine is indicated for the prevention of the novel Coronavirus infection (COVID-19) in adults aged over 18 years. 4.2. Posology and Administration: Sputnik Light vaccination consists of one dose of 0.5 ml. After the vaccine is administered, the patient should be monitored by a healthcare professional for 30 minutes. Special populations Elderly population Efficacy was similar in elderly population of more than 60 years of age as compared to adults less than 60 years of age. Pediatric population The safety and efficacy in children and adolescents (aged <18 years old) have not yet been established. No data are available. Method of administration: The vaccine is intended for intramuscular injection only. Intravenous injection of the product is strictly prohibited. The vaccine is injected into the deltoid muscle (the upper third of the outer shoulder). If it is impossible to inject into the deltoid muscle, the product is injected into the vastus lateralis muscle. For instructions on administration Prior to vaccination, thaw a vial out of the freezer and keep at room temperature (+15-25°C) until completely thawed with no visible frozen inclusions. The vial may be held in hands to help it thaw. Carefully mix the contents of the vial by swirling gently in an upright position for 10 seconds. Do not shake the vial. Remove the protective plastic cap from the vial and tear the rubber stopper with an alcohol wipe. With a single-use syringe, draw 0.5 mL of the drug as a dose to administer to the patient. After being thawed, the vaccine may be stored at room temperature (+15-25°C) for up to 2 hours. Unused contents after this period must be discarded. Refreezing of the vial with solution in not allowed. 4.3 Contraindications: • Hypersensitivity to any constituents of the vaccine. • Severe allergic reactions in the past; • Individuals below 18 years; • Acute infectious and non-infectious diseases, exacerbation of chronic diseases - vaccination is carried out 2-4 weeks after recovery or remission. In case of infectious ARVI, acute gastrointestinal infections vaccination is carried out after the temperature has returned to normal; 4.4. Use with Caution Vaccine should be used with caution in cases of chronic liver and kidney diseases, endocrine disorders (apparent thyroid function abnormalities and diabetes mellitus in decompensation stage), serious diseases of the hematopoietic system, epilepsy and other CNS diseases, acute coronary syndrome, and acute cerebrovascular event, myocardial infarction, pericarditis. Due to lack of data, vaccination may be a risk for the following groups of patients: • With autoimmune diseases (stimulation of the immune system can lead to an exacerbation of the disease, special caution should be exercised with patients with an autoimmune disorder that tend to lead to severe and life-threatening conditions); • With malignant neoplasms. The solution to vaccinate should be based on the assessment of the benefit/ risk ratio in each specific situation. 4.5. Drug Interactions No interaction studies have been performed. Concomitant administration of Sputnik Light Vector vaccine has not been studied. 4.6. Use in Special Populations (such as pregnant women, lactating women, paediatric and infertile patients, etc.) It is not anticipated that there is a biologically plausible way in which the vaccine could cause infertility in any woman or man, developmental pathology, or affect offspring, since: - The vaccine does not use adjuvants; - The potentially toxic (in ratio) excipient (polyborate 80) used in the vaccine is used in a dose that cannot affect human fertility or the reproductive function; - The vaccine virus does not reproduce itself; after injection, the virus delivers the S gene (which gene to cell to code for spike) to exist in the human body – The gene coding S protein in the body leads to the production of the viral S protein, and the development of an immune response to it - Antibodies to the S protein produced in response to immunization are similar to the antibodies produced in response to a disease caused by SARS-CoV-2, therefore, the risk associated with immunization is not higher than that for infection - Antibodies to adenovirus produced in response to immunization are similar to antibodies to adenoviruses produced in response to infection caused by adenovirus with a widely spread pathogen; therefore, the risk associated immunization is not higher than that for infection. - In preclinical studies of reproductive toxicity, a similar vaccine developed based on adenovirus vectors types 28 and 5 of identical composition was studied. No increased risk is expected with administering the drug in active reproductive populations given adherence to the restrictions indicated in the instructions for medical use. Using during pregnancy and breastfeeding There is no experience of use of Sputnik Light Vector vaccine in pregnant women, and safety and efficacy have not been established. Administration of Sputnik Light Vector vaccine in pregnancy should only be considered when the potential benefits outweigh any potential risks for the mother and foetus. Sputnik Light Vector Vaccine can be administered safely in lactating women. Pediatric Use There were no studies done on children. No children vaccination is stipulated at this development stage. Use in Elderly Subjects There were no studies done on elderly subjects. Based on the safety profile studied as part of the clinical study entitled Open Label, Tolerability, and Immunogenicity of the Sputnik Light Vector Drug, Solution for Intramuscular Injections, in Healthy Volunteers, the solution for intramuscular injections does not differ from any other similar drugs, no SAE (serious adverse events) are detected, and adverse events detected are typical for vaccines in general.																				
Specific instructions Patients undergoing immunosuppressive therapy and immunosuppressed patients may not develop a sufficient immune response. Therefore, any drugs that suppress the immune system's function are contraindicated at least within 1 month before and after vaccination due to the risk of immunogenicity reduction.																				
4.7 Effects on Ability to Drive and Use Machines There is no information regarding the effects on ability to drive and use machines.																				
4.8 Undesirable Effects Phase III Clinical Study (NCT04713480) A safety analysis in phase I-II study conducted in Russia included 110 volunteers (all volunteers were administered a dose of the study drug). A total of 181 AEs in 80 (72.73%) volunteers were recorded during this study. All the AE were non-serious. No SAEs were reported in this study. Most AEs were mild (17 AEs); 7 were moderate and one severe (hyperthermia). The most common include short-term general (a brief/flu-like syndrome characterized by chills, fever, arthralgia, myalgia, asthenia, general discomfort, headache), local (injection site tenderness, hyperemia, swelling) reactions. Non-steroidal anti-inflammatory drugs (NSAIDs) are recommended in case of post-vaccination fever and antihistamines for expressed local reactions. Common events include nausea, dyspepsia, loss of appetite, occasionally - enlarged regional lymph nodes. Some patients may develop allergic reactions, short-term elevated liver transaminase levels, elevated serum creatinine and creatine phosphokinase levels. Within the Sputnik Light vaccine, tolerability, and immunogenicity clinical trials conducted to date the following adverse reactions specific to the use of the vaccine, revealed in clinical trials and studies of other vaccines based on a similar technological platform, are predominantly mild or medium severity, and may develop during the first or second day following vaccination and usually abate within 3 common days. The common and non-common adverse reactions of short-term general (a brief/flu-like syndrome characterized by chills, fever, arthralgia, myalgia, asthenia, general discomfort, headache) or local (injection site tenderness, hyperemia, swelling) reactions. Non-steroidal anti-inflammatory drugs (NSAIDs) are recommended in case of post-vaccination fever and antihistamines for expressed local reactions. Adverse events presented below are grouped by organ systems and frequency. Frequency is defined as follows: very common (>10%), common (>100 and <10%), uncommon (>1000 and <10%), rare (>110000 and <10000), very rare (>100000, including single cases). The frequency categories have been defined based on clinical trials of the vaccine. Common events are: nausea, dyspepsia, loss of appetite, very rare – enlarged regional lymph nodes. Some patients may develop allergic reactions, short-term elevated liver transaminase levels, elevated serum creatinine and creatine phosphokinase levels.																				
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Lab test and instrumentation data: divergent deviations of immunological status indicators: increased count of lymphocytes, increase in the percentage of lymphocytes, decreased count of natural killer cells, increased count of CD4-lymphocytes, decreased count of CD4-lymphocytes, increased count of B-lymphocytes, decreased count of B-lymphocytes, increased count of natural killer cells, increased count of B-lymphocytes, increased count of immunoglobulin (IgE) in the blood, increase in the CD4/CD8 ratio, decrease in the CD4/CD8 ratio, increased level of immunoglobulin (IgA) in the blood. Abnormalities in the complete blood count: increase in the percentage of lymphocytes, decrease in the hematocrit, increased count of lymphocytes, increase in the erythrocyte sedimentation rate, increased leukocyte count, increased count of monocytes, increased platelet count, decreased count of leukocytes, decreased platelet count. Abnormalities in common urine analysis: erythrocytes in the urine. Most AEs did not cause serious reactions, with no consequences. Lab test deviations were not of clinical significance (did not require additional diagnostics or therapy).																				
4.9. Overdose Overdose cases were not reported. Considering that the dispensing of product is allowed only for medical institutions, and the vaccination itself is carried out only by qualified medical personnel, the risk of overdose is extremely low. However, it can be assumed that with an accidental overdose, the development of the above toxic and toxic-allergic reactions to a more severe degree is possible. There are no specific antidotes to the vaccine. Therapeutic measures in the case of overdose should include symptomatic therapy in accordance with the indications (antipyretic NSAIDs and desensitizing agents). Antidotes (antidotes-parentally) for severe allergic reactions. The regimen for prescribing drugs should be selected according to the recommendations for use and dosages of these products.																				
5. PHARMACOLOGICAL PROPERTIES Pharmacotherapeutic group: medical immunobiological vaccine. ATC code: J07B																				
5.1 Mechanism of action The vaccine is intended to induce the formation of humoral and cellular immunity against coronavirus infection caused by the SARS-CoV-2 virus. The mechanism of the drug's action is based on the ability of A226 based recombinant viral particles carrying the SARS-CoV-2 S protein gene to transduce efficiently the cells of the vaccinated body. In this case, genetic sequences which code the antigen is delivered to the cells, so the transduced cells start to produce the antigen.																				

When the Sputnik Light is administered (intramuscularly), the rAd26-based vector enters the cells of the body leading to the expression of SARS-CoV-2 S protein thus triggering the development of specific SARS-CoV-2 immunity.			
5.2 Pharmacodynamic properties			
Vaccine effectiveness and immunogenicity were studied in various animal models like mice, hamsters and primates. Hamster studies indicated that vaccination could achieve 100% survival in immunosuppressed hamsters when they are infected with SARS-CoV-2 virus. Primate studies indicated that there was significant immunogenicity developed in vaccinated animals in terms of glycoprotein (spike protein) specific antibodies, virus neutralizing antibody and CD4/CD8 lymphocyte proliferation.			
5.3 Immunological efficacy			
The immunological properties and safety of the vaccine have been studied in various clinical trials in adult volunteers of both sexes over the age of 18.			
An intermediate immunogenicity review revealed that the vaccine had formed an immune response in the subjects. Seroprotection was registered in 96.86% healthy volunteers on Day 28. Persons with existing immunity to coronavirus were found to have a marked growth of antibody levels on the 10th day after inoculation which may suggest an opportunity to use the drug for vaccinating people who have had COVID-19 and have a lowered antibody titer with the aim of preventing a relapse of the disease.			
A single-shot immunization in 100% healthy volunteers (all the 30 subjects whose cell mediated immunity had been studied) formed a cell immune response to SARS-CoV-2 S-protein. The protective antibody titer is currently unknown. Duration of protection is not known.			
5.4 Pharmacokinetic properties			
Target gene expression and content analysis for adenoviral DNA were evaluated in mice administered both components of the vaccine intramuscularly in thigh muscle. The gene expression peaked on day 2 to day 14 in mice organs. The adenoviral DNAs were found restricted to the thigh muscle only. No other pharmacokinetic studies were conducted with the product.			
6. NONCLINICAL PROPERTIES			
6.1 Systemic toxicity, allergenicity and immunotoxicity			
The conducted pre-clinical safety and efficacy studies of the vector vaccine against COVID-19, whose active components are recombinant adenoviral vectors, based on HAdV-26 carrying the SARS-CoV-2 virus S-protein gene, showed high safety of the vector vaccine against COVID-19 in the side range of doses, high protectivity and immunogenicity on sensitive animal species (golden hamsters).			
6.2 Carcinogenesis, Mutagenesis, Impairment of Fertility			
No such studies were conducted with the product.			
7. CLINICAL STUDIES			
Phase III Clinical Study (NCT04713488)			
A Phase III clinical study on the safety, tolerability, and immunogenicity study of "Sputnik Light" to help prevent the coronavirus infection caused by the SARS-CoV-2 virus was conducted in Russia in 110 volunteers. Immunological efficacy results demonstrates that:			
100% seroconversion in seroprotective volunteers (GMT 1648) and 92.86% seroconversion in seroprotective volunteers (GMT 10588.19) in terms of SARS-CoV-2 glycoprotein specific antibodies on Day 42 94.39% seroconversion in seroprotective volunteers (GMT 14.57) and 100% seroconversion in seroprotective volunteers (GMT 751) in terms of SARS-CoV-2 virus neutralizing antibodies on Day 42 - Strong cellular immune response in day 40 as indicated by CD4 dominant mitogen stimulated lymphoproliferation and >20 fold rise in IFN gamma induction			
Onsapon Phase III Clinical Trial (NCT0471081)			
A Phase III Clinical Trial to evaluate the efficacy, safety and tolerance of using the Sputnik Light vector vaccine in parallel groups to help prevent the infection caused by the SARS-CoV-2 infection when compared to a placebo is ongoing in Russia. Interim analysis of Phase III Clinical Trial on efficacy, safety and immunogenicity is ongoing in 4184 volunteers showed that			
- Sputnik Light vaccine provides efficacy of 82.3% against COVID-19 21 days after immunization. - A one-time immunization given using the Sputnik Light vaccine causes the formation of antigen-specific IgG antibodies to the S protein with high seroconversion of 94.7% for seroprotective individuals by day 42 after vaccination. On day 42 after vaccination, the subjects in the Sputnik Light group showed a significant increase in the titer of virus-neutralizing antibodies (NtAb) in the blood serum, with an increase from 4.3 to 78.4 (p<0.0001), i.e. 17.7 fold which indicates the formation of a specific immune response. - A single immunization given using the Sputnik Light vaccine is capable of reducing the severity of the cases involved when compared to the placebo group.			
Parameter	Sputnik Light	Placebo	
SARS-CoV-2 Glycoprotein antibodies (N = 300; 231 on Sputnik Light and 69 on placebo)			
Overall GMT at Day 42	47623 (3622, 60422.8)	23.7 (9.9, 56.3)	GMTR 201: 2 (110.2, 367.6), P < 0.0001
Overall seroconversion at Day 42	22429 (97.0%)	31/60 (44.9%)	P < 0.001
Baseline seroprotective GMT at Day 42	155.61* (1026.8,2211.1)	2.4 (1.1, 5.0)	GMTR 636.5 (326.4, 1241.2), P < 0.0001
Baseline seroprotective seroconversion at Day 42	126/133 (94.7%)	5/42 (11.9%)	P < 0.001
Baseline seroprotective GMT at Day 42	22897.7 (17691.8,29125.2)	849.9 (431.6,1673.6)	GMTR 26.71 (6.39, 43.53), P < 0.0001
Baseline seroprotective seroconversion at Day 42	98/98 (100.0%)	26/27 (96.3%)	P = 0.216
SARS-CoV-2 Neutralizing antibodies (N = 50; 39 on Sputnik Light and 11 on placebo)			
Overall GMT at Day 42	76.4 (32.7, 178.5)	4.2 (1.2, 14.3)	GMTR 18.37 (4.45, 75.91), P < 0.0001
Baseline seroprotective GMT at Day 42	8.8* (4.0, 19.4)	1.8 (0.7, 4.6)	GMTR 4.96 (1.64, 15.00), P = 0.0204
Baseline seroprotective GMT at Day 42	740.6 (75.6, 1165.4)	40.0 (1.3, 1252.2)	GMTR 18.51 (6.38, 53.73), P < 0.0001
8. DESCRIPTION			
Frozen solution. It is a dense, hardened, whitish mass. After thawing: homogeneous colorless or yellowish translucent solution.			
Characteristics: The vaccine is obtained by cDNA biotechnology methods, which does not use the SARS-CoV-2 virus pathogenic for humans. The product consists of a recombinant adenovirus (HAdV-26) with a human adenovirus serotype 28 carrying the gene for the S-protein of the SARS-CoV-2 virus.			
8.1 Pharmaceutical particulars			
8.1.1. Pharmaceutical composition:			
The product should not be mixed with any other medicinal products after active immunizing agents.			
8.1.2 Shelf-life			
Frozen drug: 6 months. Do not use after expiry date.			
Liquid drug: 1 month. Do not use after expiry date.			
8.1.3 Packaging information			
Container Closure System:			
Sputnik Light is presented as multi dose (3 mL) in each vial of vial type I (glass) with a rubber stopper and a flip-off plastic cap with aluminium seal. In total vial of multi dose contains 5 doses (3 mL).			
8.1.4 Storage and handling instructions			
Sputnik Light is stored at a temperature of -18°C or below.			
Store in a thawed state at room temperature (15-25°C			

BRAND NAME	Sputnik Light	DIMENSIONS (mm)	Open size: 105×605 ± 2 mm (L×W), Folded size: 55×52.5 ± 2 mm (L×W)
PRODUCT NAME & STRENGTH	Sputnik Light & 1.0 ± 0.5 x 10 ¹⁰ Particles / dose	SUBSTRATE	Bible Paper
CUSTOMER & COUNTRY	Hetero Biopharma & Export	GRAMMAGE (GSM)	40 GSM ± 5%
ARTWORK ITEM CODE	2068224	VARNISH	NA
SUPERSEDED CODE	Nil	COLOR CODES	Black
DESCRIPTION	Package insert printed on Bible paper on both side		
MODE OF SUPPLY (Roll Form/Bundles) Bundles			

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